SPEECH AND VIDEO PROCESSING ASSIGNMENT

Speech Emotion Recognition using the RAVDESS Dataset

Subject: Speech and Video Processing

Dataset: RAVDESS

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Section: SVP 12

Submitted as a practical study of audio feature extraction, supervised classification, and unsupervised clustering.

1. Selected Dataset

(i) Dataset link

RAVDESS emotional speech audio, Kaggle: https://www.kaggle.com/datasets/uwrfkaggler/ravdess-emotional-speech-audio

RAVDESS contains acted emotional speech and song recordings. This experiment uses audio-only speech recordings for an eight-class emotion-recognition task.

(ii) Purpose of the dataset

The experiment studies whether acoustic features can help a model identify the emotion expressed in speech. MFCC features are extracted, an SVM classifier is trained and evaluated on held-out actors, and K-Means clustering is used to explore feature similarity.

(iii) Input and output

Task	Input	Output
Emotion classification	WAV recording represented by 26 MFCC summary features	One of eight emotion labels
Clustering	Standardized MFCC vectors; labels not used for fitting	Cluster ID; clusters do not inherently equal emotion labels

Classification uses known emotion labels as targets. K-Means groups feature vectors without using emotion labels during fitting.

(iv) Number of recordings

The Kaggle download contained 2,880 WAV files. Filtering for audio-only speech (modality 03, channel 01) produced 1,440 usable recordings across 24 actors. Neutral has fewer recordings than the other emotion categories.

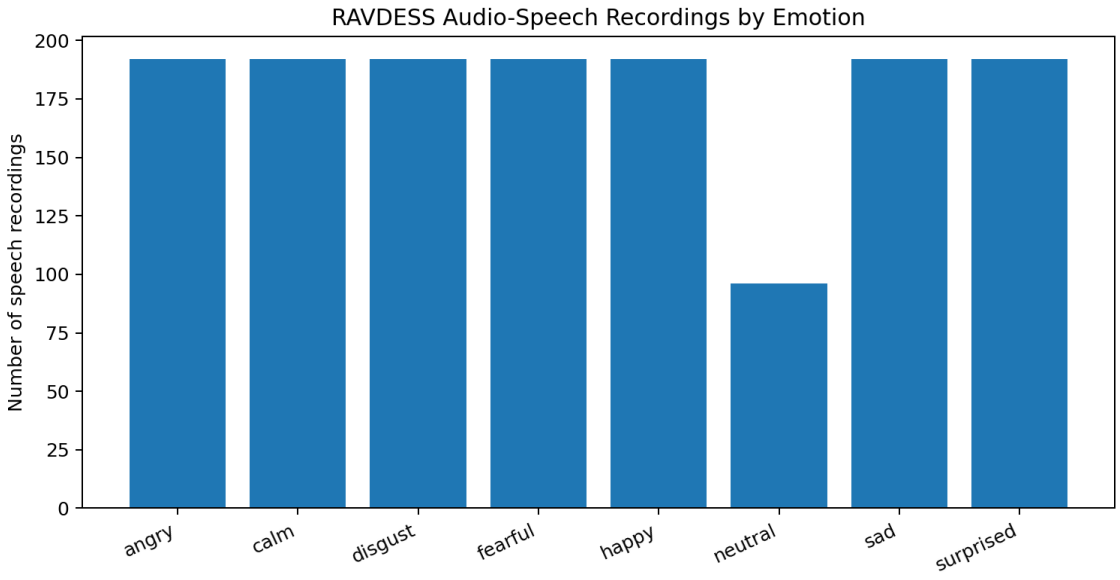


Figure 1. Emotion-class distribution for the RAVDESS audio-only speech subset.

2. Feature Extraction: MFCC and LPCC

MFCC is used in this implementation. Librosa loads each recording as mono audio at 22,050 Hz and computes 13 MFCC coefficients. The temporal mean and standard deviation of each coefficient are concatenated to form 26 features per recording. LPCC is not used; a future study could compare MFCC and LPCC with the same split and metrics.

3. Algorithms Used

(i) Support Vector Machine (SVM)

SVM is the supervised classifier. The pipeline standardizes features and uses an RBF-kernel SVC with C=10, gamma='scale', and balanced class weights. Data is split by actor to evaluate generalization to speakers not present in training.

(ii) K-Means clustering

K-Means is fit to standardized features with eight clusters. PCA provides a two-dimensional visualization. Emotion labels are used only after clustering to evaluate clusters, not to train the clustering algorithm.

(iii) Other possible algorithms

Potential future comparisons include KNN, Random Forest, Logistic Regression, DTW-based approaches, and neural networks. These were not evaluated in the reported run.

4. Experimental Workflow

• Download RAVDESS using KaggleHub.

• Parse emotion and actor information from WAV filenames.

• Filter to 1,440 audio-only speech recordings.

• Extract 13 MFCCs and summarize each by mean and standard deviation.

• Use GroupShuffleSplit with actor groups and a 25% test partition.

• Train SVM and calculate classification metrics and confusion matrix.

• Fit K-Means and evaluate with ARI and NMI.

• Save reports, plots, clustering results, and metadata to outputs.

5. SVM Classification Results

The completed run reported 39.17% test accuracy on 360 held-out recordings. The table reproduces the classification report printed by the executed script.

Emotion	Precision	Recall	F1-score	Support
angry	0.6667	0.5833	0.6222	48
calm	0.3750	0.2500	0.3000	48
disgust	0.5000	0.2708	0.3514	48
fearful	0.3671	0.6042	0.4567	48
happy	0.4848	0.3333	0.3951	48
neutral	0.2727	0.1250	0.1714	24
sad	0.2672	0.6458	0.3780	48
surprised	0.4286	0.1875	0.2609	48
Accuracy			0.3917	360
Macro avg	0.4203	0.3750	0.3670	360
Weighted avg	0.4301	0.3917	0.3800	360

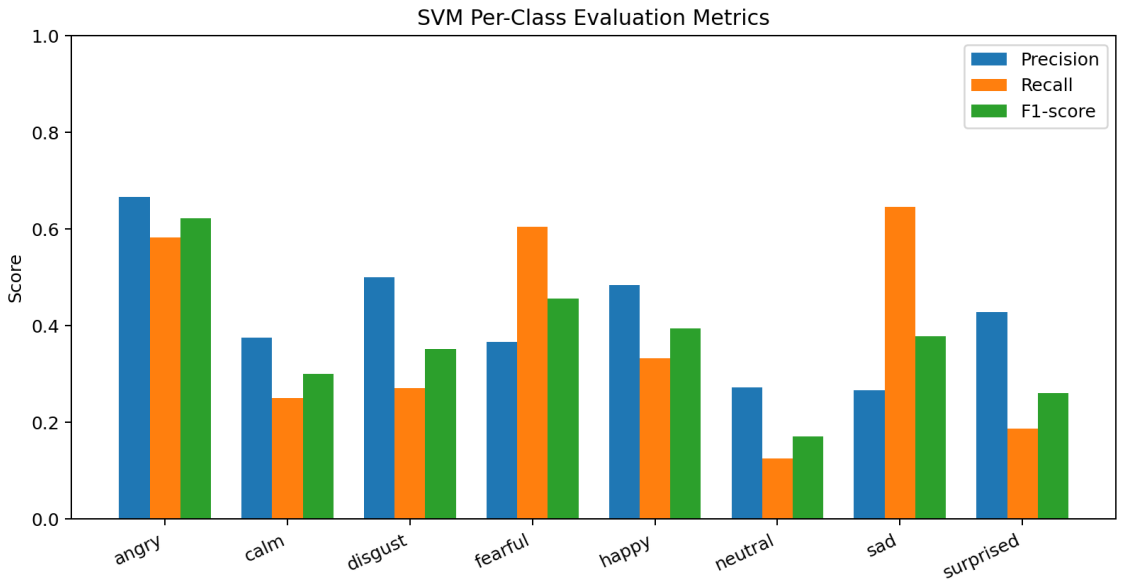


Figure 2. Per-class precision, recall, and F1-score from the actual classification report.

Interpretation of results

The highest F1-score was for angry speech (0.6222), followed by fearful speech (0.4567). Neutral speech had the lowest F1-score (0.1714); surprised speech also had a low score (0.2609). Recall was relatively high for sad (0.6458) and fearful (0.6042), but their precision values show that some predictions were incorrect.

Overall accuracy of 39.17% indicates a baseline rather than a high-performing recognizer. The 26-feature summary discards temporal patterns that may help distinguish emotion.

Confusion matrix

The executed script generated outputs/confusion_matrix.png. View or insert that image from the project outputs folder to inspect which emotion pairs are confused. Numeric cell counts are not reproduced here because they were not present in the terminal output supplied for this report.

6. K-Means Clustering

K-Means was run on standardized MFCC vectors with eight clusters. The PCA visualization is saved as outputs/kmeans_pca.png, the cluster/emotion cross-tabulation as outputs/kmeans_cluster_table.csv, and ARI/NMI as outputs/clustering_metrics.txt. Since the numeric ARI and NMI values were not included in the supplied terminal output, they are not stated here.

7. Files Generated

classification_report.txt — Accuracy and per-class precision, recall, F1-score, and support

confusion_matrix.png — Confusion matrix for the SVM test predictions

classification_metrics.png — Per-class classification metrics graph

class_distribution.png — Distribution of the speech recordings by emotion

sample_waveform.png — Waveform plot for an example speech recording

kmeans_pca.png — PCA projection of the K-Means clusters

kmeans_cluster_table.csv — Cross-tabulation of clusters against emotion labels

clustering_metrics.txt — ARI and NMI values for clustering

recordings_metadata.csv — Metadata for the selected recordings

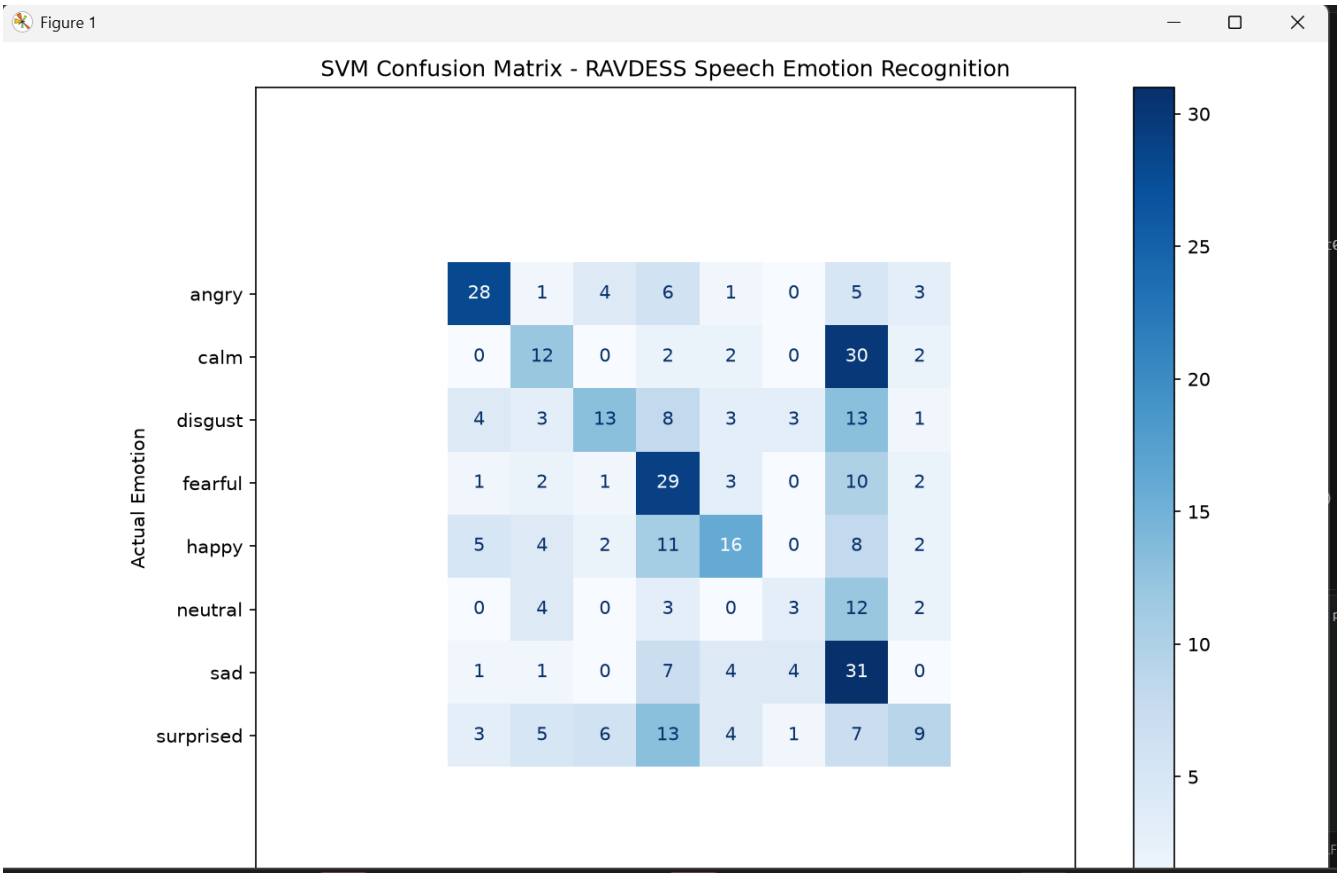
8. Limitations and Future Improvements

• Current accuracy is moderate; treat the model as a baseline.

• Try delta/delta-delta MFCCs, pitch, energy, and spectral features.

SVM CONFUSION MATRIX

Figure 3. Confusion matrix for the RAVDESS speech-emotion recognition experiment. Rows show actual emotions and columns show predicted emotions. Diagonal cells indicate correct classifications; off-diagonal cells indicate misclassifications.



Note: This is the confusion matrix image supplied after running the notebook. Confirm that it corresponds to the same model run as the classification report before submission.

• Tune SVM hyperparameters and use actor-grouped cross-validation.

• Compare SVM with KNN, Random Forest, and neural networks.

• Inspect the confusion matrix before drawing conclusions about emotion pairs.

• Add actual ARI/NMI values from clustering_metrics.txt.

• Acted speech may not generalize to spontaneous real-world speech.

9. Conclusion

This assignment implemented an end-to-end speech-emotion-recognition pipeline on RAVDESS audio-only speech. The script selected 1,440 recordings, extracted 26 MFCC summary features, trained an SVM, and evaluated it on 360 recordings held out by actor. Accuracy was 39.17%, with the strongest F1-score for angry speech and the weakest for neutral speech. K-Means was also used to explore the acoustic feature space. The experiment demonstrates feature extraction and model evaluation while identifying opportunities for improvement.

10. References

RAVDESS emotional speech audio dataset, Kaggle: https://www.kaggle.com/datasets/uwrfkaggler/ravdess-emotional-speech-audio

Librosa MFCC documentation: https://librosa.org/doc/latest/generated/librosa.feature.mfcc.html

Scikit-learn Support Vector Machines: https://scikit-learn.org/stable/modules/svm.html

Scikit-learn clustering: https://scikit-learn.org/stable/modules/clustering.html

Scikit-learn classification report: https://scikit-learn.org/stable/modules/generated/sklearn.metrics.classification_report.html

Scikit-learn GroupShuffleSplit: https://scikit-learn.org/stable/modules/generated/sklearn.model_selection.GroupShuffleSplit.html